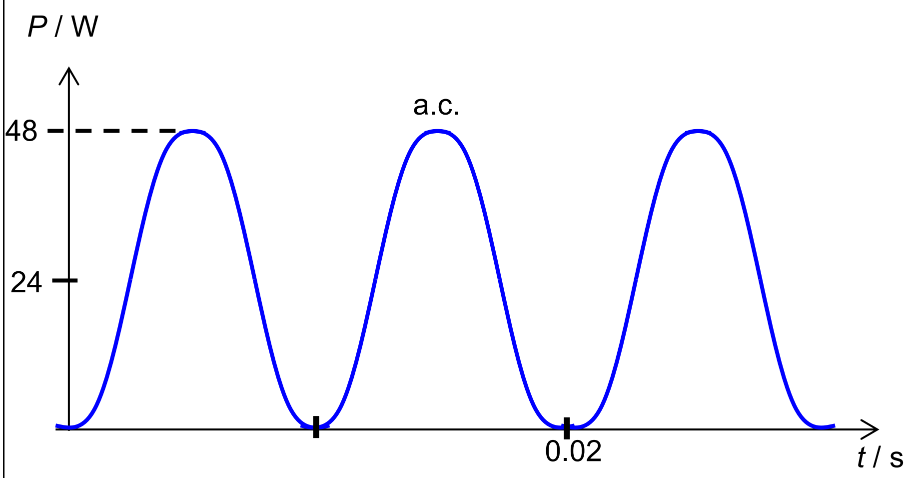
Questions			Answers	Marks
4	**(a)**		Resistivity (ρ) is the proportionality constant between the dimensions of a specimen of a material and its resistance (that is constant at constant temperature) such that $R = \dfrac{\rho L}{A}$	B1
	(b)	**(i)**	N.B. If the soil is acidic, it reacts with copper and this produces an e.m.f. This occurs due to a galvanic reaction, where two dissimilar metals (in this case, copper plates) in an electrolyte (the acidic soil) create a voltage. V = 1.398 – 0.281 = 1.117 V I = 0.31x10^{-3} A R = V/I = 3603 ~ 3600 Ω	M1
		(ii)	R = $\rho l/A$, thus ρ = RA/l = 3600 x 0.800 x 0.210 / 0.900 = 670 Ω m (2 s.f.)	M1 A1
		(iii)	Any one of these:	

			• Among all the readings given, the least significant or most imprecise is the current reading (only 2 s.f.). The current reading will be subject to significant random errors. EITHER: Increase the area of the copper plates **in the soil** OR descrease the distance between copper plates. This will decrease the resistance of the sample of soil to be measured and increase the current readings for the same voltage applied. Main point is to increase measured current, so that it will be more than 2 s.f., as given in the question.	M1 A1
	(c)	**(i)**	Power = $\frac{V^2}{R} = \frac{12^2}{6.0} = 24$ W	A1
		(ii)	 1 mark for correct graph (at least two periods) for ac supply 1 mark for labelling of values on axes	B2
		(iii)	Power input = Power output $0.7V_1 I_1 = V_2 I_2$ $0.7(230) I_1 = 24$ $I_1 = 0.15$ A	M1 A1
		(iv)	Voltage can be easily <u>stepped up or down</u> using a transformer in order to reduce power loss in transmission wires.	B1